
PROJECT PROPOSAL

INTERNSHIP MANAGEMENT PORTAL

A Full-Stack Web Application for Digitizing
Internship Operations at IED India Pvt Ltd



Project Proposal — Internship Management Portal

1. Name of the Company

IED India Pvt Ltd is a multifaceted organization headquartered in India, dedicated to empowering future leaders through structured internship and training programmes. The company operates across six core departments — Digital Marketing, HR & Recruitment, Business Development, Social Media, Entrepreneurship Training, and IT Support — and hosts a substantial number of interns throughout the year. Each department runs its own internship track, requiring coordinated management of applications, interviews, attendance, task assignments, performance evaluations, and certificate issuance across the organization.

2. Objectives of the Internship Study

The primary objectives of developing the Internship Management Portal are:

- To conduct a thorough analysis of the existing manual internship management processes at IED India Pvt Ltd and identify key operational inefficiencies, bottlenecks, and data inconsistencies.
- To design and develop a centralized, full-stack web application that automates the complete internship lifecycle — from the initial application submission through interview scheduling, candidate selection, onboarding, daily attendance tracking, task management, weekly performance evaluations, and final certificate generation.
- To implement a robust Role-Based Access Control (RBAC) system supporting four distinct user roles: Admin, HR, Mentor, and Intern, each with a tailored dashboard, navigation, and permission set ensuring both data security and operational efficiency.
- To integrate automated email notification capabilities for all critical events including application confirmations, interview scheduling with calendar details and meeting links, status updates, and welcome messages upon selection.
- To build a digital certificate generation system capable of producing professionally designed PDF certificates with embedded QR codes for instant online verification, eliminating the need for manual document preparation.
- To deploy the application on Vercel's serverless infrastructure with MongoDB Atlas as the cloud database, ensuring high availability, automatic scaling, zero-maintenance hosting, and continuous deployment from the GitHub repository.

3. Scope of the Study

The scope of this study encompasses the complete design, development, testing, and deployment of a production-grade web application comprising 11 functional modules:

- Application Management — A public-facing internship application form with document uploads (resume, photo), followed by an admin dashboard for filtering, reviewing, shortlisting, and processing applications through a six-stage status workflow (Applied, Shortlisted, Interview Scheduled, Selected, Rejected, On Hold).
- Interview Scheduling — Multi-mode interview management (Zoom, Google Meet, Offline, Phone) with automated email notifications containing meeting links, date/time details, and post-interview feedback with scoring.
- Onboarding — A structured checklist tracking offer letter dispatch, orientation scheduling, document verification (Aadhaar, college ID, resume, photo), agreement upload, and welcome email delivery.
- Attendance Tracking — Daily check-in/check-out with optional GPS geo-location capture, supporting Office, Work From Home, and Leave attendance types with monthly calendar views and statistics.
- Task Management — Assignment, tracking, and review of tasks with deadlines, priority levels, file attachments, submission workflows, mentor feedback, and completion scoring.
- Communication — Direct messaging between users with file attachments and read receipts, plus organization-wide announcements with role targeting and pin functionality.
- Performance Evaluation — Structured weekly evaluations rating interns across six dimensions (Communication, Teamwork, Leadership, Discipline, Technical Skills, Task Completion) with auto-calculated scores and recommendations.
- Leave Management, Certificate Generation, and Analytics Dashboard — Leave request workflows, PDF certificate generation with QR verification, and real-time analytics with Chart.js visualizations.

The system serves all internal stakeholders — administrators, HR personnel, mentors, and interns — through a single, unified, responsive web interface accessible on both desktop and mobile devices.

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4. Methodology

The project follows the Agile Software Development methodology, employing iterative development cycles with continuous testing and refinement. The system architecture adheres to the three-tier pattern with a clear separation between the Presentation Layer (client), the Application Layer (server), and the Data Layer (database).

The backend is built using Node.js as the runtime environment and Express.js (v4.19) as the web framework, exposing a comprehensive RESTful API with over 50 endpoints across 13 route groups. MongoDB (v7.x) serves as the NoSQL database, accessed through the Mongoose ODM (v8.4) which provides schema-based modelling with built-in validation, indexing, and population (joins). Security is enforced through multiple layers: JSON Web Token (JWT) based stateless authentication with configurable expiry, bcryptjs password hashing with 12 salt rounds, Helmet.js for HTTP security headers (CSP, HSTS, X-Frame-Options), and role-based authorization middleware.

The frontend is implemented as a Single Page Application (SPA) using vanilla HTML5, CSS3, and JavaScript (ES2020+), avoiding heavy framework dependencies. The UI features a modern dark-mode glassmorphic design with a custom CSS component library (25,500+ bytes) providing reusable status badges, modal dialogs, toast notifications, and responsive data tables. Lucide Icons provides the icon set, and Chart.js (v4.4) powers the interactive analytics visualizations. PDFKit (v0.15) handles server-side certificate PDF generation with in-memory buffers, while the QRCode library generates embedded verification QR codes.

5. Data Collection

The data collection methodology combines both primary and secondary research approaches:

- **Primary Data:** Requirements were gathered through direct discussions and interviews with IED India's management team, HR department, and mentors to understand the current manual workflows. Observation of the existing processes — which relied on spreadsheets, email threads, WhatsApp groups, and physical records — revealed critical pain points including data fragmentation, delayed status communication, inconsistent attendance records, and time-consuming certificate preparation.
- **Secondary Data:** Technical best practices and design patterns were sourced from the official documentation of Node.js, Express.js, MongoDB, Mongoose, Vercel, and JWT. Industry research on internship management systems, HR technology platforms, and SaaS application design informed the feature set and user experience decisions. Security standards from OWASP guidelines were referenced for implementing authentication, authorization, and input validation.

6. Plan of Analysis

The analysis evaluates the effectiveness of the developed system across three critical dimensions:

- **Operational Efficiency:** Measuring the quantitative reduction in administrative time and effort for key processes — application processing (from manual email sorting to automated filtering and status workflows), interview scheduling (from back-and-forth emails to one-click scheduling with auto-notifications), attendance tracking (from manual spreadsheet entry to digital check-in/check-out with geo-verification), and certificate generation (from manual Word document creation to instant PDF generation).
- **Data Integrity & Reliability:** Verifying the consistency and accuracy of data across all 12 MongoDB collections through referential integrity checks (ObjectId references between Users, Applications, Tasks, Evaluations), compound unique indexing (Attendance: internId + date), schema-level validation (enum constraints, required fields, min/max values), and the custom toJSON transformation that excludes sensitive fields like passwords from API responses.
- **User Experience & Accessibility:** Assessing the usability and responsiveness of the interface through multi-role dashboard testing (Admin, HR, Mentor, Intern), cross-browser compatibility verification (Chrome, Firefox, Edge, Safari), responsive design validation across desktop (1920px), tablet (768px), and mobile (375px) viewports, and real-time notification delivery testing across 50+ API endpoints.

7. Chapter Scheme

The project report is organized into the following seven chapters:

Chapter 1: Introduction

Provides the background context, problem statement, project objectives, and an overview of the scope and significance of the study.

Chapter 2: Industry & Company Profile

Presents a detailed profile of IED India Pvt Ltd, its organizational structure, departmental operations, internship programme history, and the specific challenges that motivated this project.

Chapter 3: Literature Review

Reviews existing internship management systems, HR technology platforms, and relevant academic literature on web application development, SPA architecture, and NoSQL database design.

Chapter 4: System Design

Documents the system architecture, technology stack selection rationale, database schema design (12 collections), API endpoint structure (50+ endpoints), and user interface wireframes.

Chapter 5: Implementation

Details the development process including the MVC code structure, authentication flow, role-based access control logic, email service integration, certificate generation pipeline, and serverless deployment configuration.

Chapter 6: Testing & Results

Presents the functional testing results, integration testing outcomes, cross-browser compatibility findings, and performance benchmarks demonstrating the system's effectiveness.

Chapter 7: Conclusion & Future Scope

Summarizes the project achievements, lessons learned, and outlines planned enhancements including WebSocket notifications, mobile applications, AI-powered task grading, and custom domain email integration.